

DS Unit 2 Previous Year Questions

Data Structures (DS) · Unit 2 · Stacks & Queues · 2023

Questions collected from previous year university examinations (2019–2023).

Part A — Short answer (2 marks)

1. Illustrate Stack ADT with a worked example. [2021]
2. Explain Infix to Postfix Conversion with a neat diagram and a suitable example. [2020]
3. Illustrate Recursion and the Call Stack with a worked example. [2023]
4. Compare Queue ADT with Infix to Postfix Conversion. When would you prefer one over the other? [2020]

Part B — Long answer (10 marks)

1. Explain Deque and Priority Queue with a neat diagram and a suitable example. [2023]
2. Illustrate Applications of Stacks and Queues with a worked example. [2019]
3. Compare Stack ADT with Deque and Priority Queue. When would you prefer one over the other? [2020]
4. Compare Infix to Postfix Conversion with Applications of Stacks and Queues. When would you prefer one over the other? [2022]
5. Define Recursion and the Call Stack. Derive or justify the key properties involved. [2020]
6. Define Queue ADT. Derive or justify the key properties involved. [2022]
7. What is meant by Deque and Priority Queue? Discuss its advantages and limitations. [2021]
8. Compare Applications of Stacks and Queues with Stack ADT. When would you prefer one over the other? [2023]
9. Write short notes on Stack ADT. [2021]
10. Write short notes on Infix to Postfix Conversion. [2020]